Time translation symmetry may be the most fundamental symmetry of all.

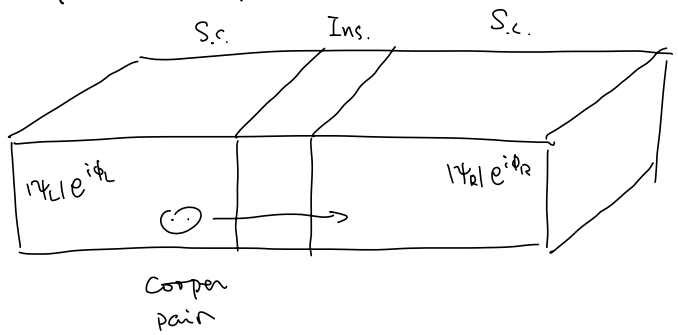
TTS is connected through Noether's theorem, to the conservation of energy.

Spontaneous breaking of spatial translation symmetry $\Longrightarrow$ solid

physicists time? time crystal?

△
- Josephson time crystals
- Floquet time crystals
- pre-thermal time crystals
- open time crystals

* AC Josephson effect



$$L = \hbar n \dot{\delta} - \frac{\hbar j_0}{2e} \cos\delta - 2neV$$

phase/number conjugacy

misalignment energy

electric potential energy

$$j = j_0 \sin\delta \qquad \text{DC Josephson}$$

$$\frac{d\delta}{dt} = \frac{2eV}{\hbar} \qquad \text{"Schrödinger"}$$

$$j(t) = j_0 \sin\left(\frac{2eV}{\hbar} t + \delta_0\right)$$

TTSB

NG mode

* Floquet time crystals

There is a periodic driving term, but the responce has a longer (doubled, tripped, ...) period.

$\Longrightarrow$ An violation of discrete version of TTS spontaneously.

interaction $\Longrightarrow$ desire to synchronize

disorder $\Longrightarrow$ ability to synchronize

A Floquet system, evolves according to a time-dependent, local Hamiltonian with $H(t+T) = H(t)$.

The unitary evolution operator $U(T)$ is diagonalized by "Floquet eigenstates". They are periodic up to a phase, and observables within them are simply periodic.

$$\psi_j(t+T) = e^{i\epsilon_j T}\psi_j(t)$$

A complete set of Floquet eigenstates always exists. But they might not be physically reasonable states. They might involve coherent superpositions of macroscopically distinct pieces (Cat state). Such states do not arise in practice.

Physically reasonable states satisfy the so-called "cluster decomposition property":

$$\langle \Phi | O_1(x)\, O_2(y) | \Phi \rangle \longrightarrow \langle \Phi | O_1(x) | \Phi \rangle \langle \Phi | O_2(y) | \Phi \rangle$$

as $|x-y| \to \infty$

for all local observables.

$|UP\rangle = |\cdots\uparrow\uparrow\uparrow\cdots\rangle$, $|DOWN\rangle = |\cdots\downarrow\downarrow\downarrow\cdots\rangle$

satisfy cluster decomposition property , but

$\frac{1}{\sqrt{2}}\left(|UP\rangle \pm |DOWN\rangle\right)$ does not.

The basic Floquet time crystal is simple :

① H_{MBL} eigenstates are locally ordered w.r.t. the z-direction of spin

② One flips all these spin.

e.g. $H_{MBL} = H_1 + H_2 = \sum_{i,j} J_{ij}\sigma_i^z\sigma_j^z + \sum_i D_i\sigma_i^z$

$H_3 = g(1-\epsilon)\sum_i \sigma_i^x$

The eigenstates of this total evolution are coherent sums of an H_{MBL} eigenstate z-spin configuration and its flipped version.

For a robust class of Hamiltonian H_{MBL}, those two states prove to be macroscopically distinct.

In that case, the Floquet eigenstates do not obey cluster decomposition. They are unphysical, cat states.

For physically realizable states, symmetry under time translation through one cycle period is broken.

On the other hand, time translation symmetry through two cycle periods remains valid, so we have period doubling.

* pre-thermal time crystals

Under broader circumstances, high-frequency drives exhibit a very-long-lived intermediate time crystal phase, though they eventually reach complete equilibrium.

Theoretically, pre-thermal time crystals can break continuous TTS.
In these examples one has $e^{-iHT} \sim I$

numerically ✓
experiment ✓

* Open time crystals

The kind of closed systems that lead to pre-thermal time crystals, when coupled to a cold bath, form another class, the "open" time crystals.

numerically ✓
experiment ✓

- What are the thermal and electrical properties of these new states of matter?
  what are their quasiparticles?
- Where and when will pre-thermal and open time crystals be observed?
- Can one characterize the various phase transition?
- Are there time liquid?
- Are there time quasi-crystals?
- Can one classify - and realize - spacetime crystals?
  Can one study their defects and topology?

time independent Hamiltonian H

$$\langle \Psi_g | \mathcal{O}(x,t) | \Psi_g \rangle = f(t)$$

$$\langle \Psi_g | \mathcal{O}(x,t)\, \mathcal{O}(0,0) | \Psi_g \rangle = f(t)$$

$$\mathrm{Tr}\left[\mathcal{O}(x,t)\, \mathcal{O}(0,0)\, e^{-\beta H} \right] = f(t)$$

<u>No-go theorems</u> 2015, Watanabe, Oshikawa

$\Longrightarrow$ periodic driving $\quad H(t+T) = H(t) \quad$ Discrete TTSB

<u>Def</u> A system in a many-body steady state for a periodically driven Hamiltonian with period T is a discrete time crystal if the expectation values of local observables are not T-periodic.

arXiv 2302.08056

An analytical theory of spectral pairing for mesoscopic clean DTC

MBL : SP occurs for all eigenstates $\Longrightarrow$ any initial state exhibits local DTC oscillation.

Prethermal : initial states belonging to low-temperature sectors $\begin{cases} \text{eigenstate SP} \\ \text{DTC dynamics} \end{cases}$

<u>Model</u> $\quad H_0(t+T) = H_0(t)$, $H(t) = H_0(t) + \lambda H'$

↳ perturbation

$$\frac{H_0(t)\,T}{2\hbar} = \begin{cases} \pi/2 \sum_{j=1}^{L} \sigma_j^x & t \in [0, \frac{T}{2}) \\ \sum_{j=1}^{L} J_j \sigma_j^z \sigma_{j+1}^z & t \in [\frac{T}{2}, T) \end{cases}$$

$$\frac{H'\,T}{2\hbar} = \sum_{j=1}^{L} \left(\phi\, \sigma_j^x \sigma_{j+1}^x + \sum_{\mu=x,y,z} \theta_\mu \sigma_j^\mu \right) , \quad \phi^2 + \sum_{\mu=x,y,z} \theta_\mu^2 = 1$$

* $\lambda = 0$: spins are perfectly flipped periodically

$$\sigma_j^z(nT) = (U_F^\dagger)^n \sigma_j^z U_F^n = (-1)^n \sigma_j^z$$

$$U_F = \mathcal{T} e^{-i/\hbar \int_0^T H(t)\,dt} \quad \text{Floquet operator}$$

Observable : spatiotemporal correlator

$$M(nT) = \frac{(-1)^n}{L} \sum_{j=1}^{L} \langle \psi_{ini} | \sigma_j^z(nT) | \psi_{ini} \rangle \langle \psi_{ini} | \sigma_j^z | \psi_{ini} \rangle$$

SP: Floquet eigen-solutions $U_F|\omega_n\rangle = e^{i\omega_n}|\omega_n\rangle$ satisfying two conditions.

① Pairwise Fock space localization

a pair of eigenstates $|\omega_1\rangle$, $|\omega_2\rangle$ are dominated by different linear combinations of just two Fock product states $|\{S_j\}_1\rangle$, $|\{S_j\}_2\rangle$.

$$|\{S_j\}\rangle \equiv |S_1\rangle \otimes |S_2\rangle \otimes \cdots \otimes |S_L\rangle \quad , \quad \sigma_j^z |S_j\rangle = \underset{\hookrightarrow \pm 1}{S_j} |S_j\rangle$$

② Fixed spectral gap

The frequency difference $\Delta\omega = \omega_1 - \omega_2$ of the eigenstate pair remains unchanged under generic perturbations.

<u>Symmetry indicators</u>

At $\lambda = 0$, $U_0 \equiv U_F(\lambda=0) = (-i)^L \prod_j e^{-i\sum_j J_j \sigma_j^z \sigma_{j+1}^z} P$

$$P = \prod_{j=1}^{L} \sigma_j^x \quad \text{flips all spin}$$

Solutions to $U_0|l, \{S_j\}\rangle = e^{iE(l,\{S_j\})}|l, \{S_j\}\rangle$ then read

$$|l, \{S_j\}\rangle = \sum_{m=0,1} (-1)^{ml} |(-1)^m \{S_j\}\rangle / \sqrt{2} \qquad l = 0, 1 \mod 2$$

$$E(l, \{S_j\}) = E_{sp}(l) + E_{Ising}(\{S_j\}) \mod 2\pi$$

$$E_{sp}(l) = \pi l \quad , \quad E_{Ising}(\{S_j\}) = -\sum_j J_j S_j S_{j+1}$$

$|0, \{S_j\}\rangle$, $|1, \{S_j\}\rangle$ may be thought to satisfy SP, as they consist of two Fock states $|\pm\{S_j\}\rangle$ and differ in quasi-energy by $E(1, \{S_j\}) - E(0, \{S_j\}) = \pi$.

$$E_{Ising} = -J(L - 2w) \qquad w = \sum_{j=1}^{L} (1 - S_j S_{j+1})/2 \quad \text{number of DW}$$

$\Longrightarrow$ Majority eigenstates are grouped into degenerate subspace labelled by (l, w), each spanned by large numbers of configurations $\{|l, \{S_j\}_1\rangle, |l, \{S_j\}_2\rangle \cdots\}$

with different allocations of ω DWs.

To identify non-degenerate scars efficiently : symmetry-based algorithm

T_x : spatial translation symmetry $\quad [T_x, H(t)] = 0$

$$T_x |\{S_j\}\rangle = T_x |S_1 S_2 \cdots S_{L-1} S_L\rangle = |S_L S_1 \cdots S_{L-1}\rangle = |\{S_{j-1}\}\rangle$$

* Scar configurations should exhibit identical DW numbers at all symmetry related bonds, s.t. relocations of DWs cannot produce new degenerate configurations.
* $|\pm\{S_j\}\rangle$ host identical DW distributions.

$\Longrightarrow$ Scar pattern only need to satisfy a projective translation symmetry

$$T_x |\{S_j\}\rangle = |\pm\{S_j\}\rangle$$

$\pm$: give the FM, AFM cat scars

$$|\ell, FM\rangle \equiv |\ell, \{S_j = (+1)^j\}\rangle \quad , \quad |\ell, AFM\rangle \equiv |\ell, \{S_j = (-1)^j\}\rangle$$

$|\pm\{S_j\}\rangle$ are recombined into non-degenerate spectral pair satisfying

$$P|\ell, \{S_j\}\rangle = (-1)^\ell |\ell, \{S_j\}\rangle$$

PRL 129. 133001

DTC Enforced by Floquet-Bloch Scars

Bosons system, $\hat{H}(t+T) = \hat{H}(t)$

$$\begin{cases} \dfrac{\hat{H}_1 T}{2\hbar} = \phi_1 \sum\limits_{\vec{r},\, \mu \neq \nu} i f_{\mu\nu} \left[\hat{\psi}^{+}_{\vec{r}\mu} \hat{\psi}_{\vec{r}\nu} + \lambda \hat{\psi}^{+}_{\vec{r}+\vec{e}_\mu,\, \mu} \hat{\psi}_{\vec{r}+\vec{e}_\nu,\, \nu} \right], & t \in [0, \tfrac{T}{2}) \\ \dfrac{\hat{H}_2 T}{2\hbar} = \sum\limits_{\vec{r},\, \mu} \left[\phi_2 \hat{n}_{\vec{r}\mu} (\hat{n}_{\vec{r}\mu} - 1) + \theta_\mu \hat{n}_{\vec{r}\mu} \right], & t \in [\tfrac{T}{2}, T) \end{cases}$$